
DADS 6005

Data Streaming and Real Time Analytics

Lab1: big data review

Outline

- VM setup
- Hadoop
- Mapreduce
- Sqoop
- Flume

Quickstart Virtual Machine Installation

1. Install Oracle VirtualBox (Virtual Machine) on Window
(Requires RAM at least 8 MB)
 - 1.1 Download VirtualBox from
<https://www.virtualbox.org/wiki/Downloads>
 - 1.2 Install VirtualBox by executing the downloaded file.
2. Download the zip file of the Cloudera Quickstart Machine Image from the MS Teams. Then unzip the file (using 7-zip).
3. Start the VirtualBox and click File->Import Appliance then choose the unzip file (.vmdk) from 2. to import the cloudera virtual machine.
4. When the Quickstart completely starts, open a terminal window (click on a terminal icon) to work with the machine.

Hadoop Lab 1

1. Under Quickstart
2. `# cd /home/pok`
3. Edit a file, “test.txt”, using nano command as follows
`# nano`

Enter multiple words and multiple lines, then press ctrl o to save to the file named test.txt, and then ctrl x to exit from the nano command. The file will be used for word count problem later.

Hadoop Lab 1

5. Execute the following hadoop file system commands.

```
# hadoop fs -put test.txt /home/cloudera/data/test.txt
# hadoop fs -ls /home/cloudera/data
# hadoop fs -cat /home/cloudera/data/test.txt
# hadoop fs -get /home/cloudera/data/test.txt test1.txt
# hadoop fs -mkdir /home/cloudera/temp
# hadoop fs -cp /home/cloudera/data/test.txt
    /home/cloudera/temp/test.txt
# hadoop fs -rm /home/cloudera/temp/test.txt
# hadoop fs -rmdir /home/cloudera/temp
# hadoop fs -ls
```

Hadoop Lab 1

Submit a MapReduce job for Word Count problem

- Submit a MapReduce job through Hadoop Streaming

```
# hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -input  
/home/cloudera/data/test.txt -output /home/cloudera/data/wc -  
mapper 'python mapper.py' -reducer 'python reducer.py' -file  
mapper.py -file reducer.py
```

Output will be written in multiple files under /home/cloudera/wc e.g.
part-00000,

Edit a mapper.py as follows:

```
#!/usr/bin/env python
```

```
from operator import itemgetter  
import sys
```

```
for line in sys.stdin:  
    for word in line.split():  
        print(word+"\t1")
```

Edit a reducer.py as follows:

```
#!/usr/bin/env python

from operator import itemgetter
import sys

curkey = None
total = 0
for line in sys.stdin:
    key, val = line.split("\t", 1)
    val = int(val)

    if key == curkey:
        total += val
    else:
        if curkey is not None:
            print(curkey+"\t"+str(total))
        curkey = key
        total = val

print(curkey+"\t"+str(total))
```